Purity of Pure Is Refuted by a Universal Proposition

A hand proof of Jacob Lettie's theorem $\mathrm{W} \vdash {\sim}\mathrm{PP}_u$

11 November 2026

The theorem and its proof are due to Jacob Lettie, working with the Purity of Pure team, a group of Claude Opus 5.5 sessions (*If a Proposition Is Fundamental, Purity Is Not Pure* and *Weaker Theories That Still Refute Purity of Pure*, 9 November 2026). They are included in this repository with Lettie's permission. This note restates the result in Goodman's notation and gives a self-contained proof in our own words. The theorem is kernel-checked as `w_not_pp`, and the definitions below match that formal statement exactly. Each lemma names, in brackets, the Isabelle theorem that states it; the full table is in `docs/LETTIE_PP_CORRESPONDENCE.md`.

Setting

Propositions have type t, and $u = t \to t$. We write π for the constant Pure_u, so that PP_u is $\mathrm{Pure}_{u\to t}(\pi)$. As usual $\Box p := (p = \top)$ and $\Diamond p := {\sim}\Box{\sim}p$.

The base logic is H + PE: the system H of classical higher-order logic with $\beta\eta$-conversion and Leibniz identity, together with the Rule of Propositional Equivalence (PE): if $\vdash A \leftrightarrow B$ then $\vdash A = B$. Modalized Functionality is not assumed. PE applies only to theorems of H + PE, in the full language including the constants Pure_σ, and never to consequences of the premises below. We use these facts about H + PE:

(N) Necessitation: if $\vdash A$ then $\vdash \Box A$, for formulas A with or without free variables.

(S4) The schemas T ($\Box A \to A$), K and 4 ($\Box A \to \Box\Box A$).

(I) $\Box(p \leftrightarrow q) \to p = q$, and Leibniz's law: if $a = b$ then $Z(a) = Z(b)$ for every Z.

(B) Boolean identities from PE, for instance ${\sim}{\sim}x = x$ and $(A \,\&\, B) = (B \,\&\, A)$.

Each is a standard consequence of H + PE; see Lettie's main note, §5.

Definitions

All of the following are terms built from logical vocabulary and the constant π. The variables x, y, z, s range over propositions and Y over operators of type u.

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\begin{aligned}
R(x,y) &:= \forall Y\,(\pi(Y) \to (Y(y) \to \Diamond Y(x))) && x \text{ can have every pure property of } y,\\
\mathrm{Uni}(x) &:= \forall y\, R(x,y) && x \text{ is universal},\\
F(z) &:= \Box(\sim R(z,\sim z)\,\&\,\sim R(\sim z,z)) && z \text{ is firm},\\
Q(z) &:= F(z) \to z,\\
P(z) &:= \sim z\,\&\,\sim F(z), \qquad P'(z) := z\,\&\,\sim F(z).
\end{aligned}
$$

For an operator Y write $Y \circ Q$ for $\lambda z.\, Y(Q(z))$. Abstracting π gives closed logical terms M and M' of type $(u \to t) \to u$, and M'' of type $(u \to t) \to (u \to u)$, with

$$M(\pi) = P, \qquad M'(\pi) = P', \qquad M''(\pi) = \lambda Y.\, Y \circ Q$$

by β-conversion.

The theory W

For types σ, τ let $\mathrm{App}(\sigma,\tau) := \forall X^{\sigma\to\tau} \forall y^{\sigma}\,((\mathrm{Pure}(X)\&\mathrm{Pure}(y)) \to \mathrm{Pure}(X(y)))$. The theory W has base H + PE and exactly ten premises:

(W1–3) $\mathrm{Pure}(M)$, $\mathrm{Pure}(M')$, $\mathrm{Pure}(M'')$;

(W4–7) $\mathrm{App}(u \to t, u)$, $\mathrm{App}(u \to t, u \to u)$, $\mathrm{App}(u, t)$, $\mathrm{App}(u, u)$;

(W8–9) $\mathrm{Pure}_t(\pi(P)) \to (\pi(P) \to \Box\pi(P))$ and the same for P';

(W10) $\exists x\, \mathrm{Uni}(x)$.

W1–W7 are instances of Logical Purity and Application, and W8–W9 are instances of zeroary Exhaustion. All ten are ordinary premises: none is assumed to be necessary, and Necessitation is never applied to a formula whose proof uses them.

Theorem (Lettie; `w_not_pp`)**.** $\mathrm{W} \vdash \sim\mathrm{PP}_u$. *Equivalently,* PP_u *and W1–W9 prove* $\sim\exists x\, \mathrm{Uni}(x)$*: no proposition is universal.*

Proof

Lemmas A–F are theorems of H + PE (they mention π but use no premise), so (N) may be applied to them.

Lemma A (`nup_lemA`)**.** $F(x) \to \Box F(x)$.

Proof. $F(x)$ is a boxed proposition, so this is an instance of axiom 4. □

Lemma B (`nup_lemB`)**.** $F(x) \to Q(x) = x$.

Proof. The formula $F(x) \to (Q(x) \leftrightarrow x)$ is a tautology, since $Q(x)$ is $F(x) \to x$. By (N) and K, $\Box F(x) \to \Box(Q(x) \leftrightarrow x)$. Assume $F(x)$. Lemma A gives $\Box F(x)$, hence $\Box(Q(x) \leftrightarrow x)$, hence $Q(x) = x$ by (I). □

Lemma C (`nup_lemC, nup_ident`)**.** $F(\sim x) = F(x)$. *Consequently* $P'(z) = P(\sim z)$.

Proof. Write $C(x)$ for $\sim R(x, \sim x)\ \&\ \sim R(\sim x, x)$, so that $F(x) = \Box C(x)$. Since $\sim\sim x = x$ by (B), Leibniz's law turns $C(\sim x)$, which is $\sim R(\sim x, \sim\sim x)\ \&\ \sim R(\sim\sim x, \sim x)$, into $\sim R(\sim x, x)\ \&\ \sim R(x, \sim x)$. This is $C(x)$ with its conjuncts exchanged, so it equals $C(x)$ by (B). Hence $C(\sim x) = C(x)$, and Leibniz's law with $Z := \lambda p.\Box p$ gives $F(\sim x) = F(x)$.

For the second claim, $P(\sim z)$ is $\sim\sim z\ \&\ \sim F(\sim z)$. Rewriting $\sim\sim z$ to z and $F(\sim z)$ to $F(z)$ by Leibniz's law gives $z\ \&\ \sim F(z)$, which is $P'(z)$. □

Lemma D (`nup_lemD`)**.** $\sim Q(x) \to F(Q(x))$.

Proof. Assume $\sim Q(x)$, that is, $\sim(F(x) \to x)$. Then $F(x)$. Lemma B gives $Q(x) = x$, so $F(Q(x)) = F(x)$ by Leibniz's law, and $F(Q(x))$ holds. □

Lemma E (`nup_cor`)**.** $\Box \sim P(Q(x))$.

Proof. Suppose $P(Q(x))$, that is, $\sim Q(x)\ \&\ \sim F(Q(x))$. Lemma D turns the first conjunct into $F(Q(x))$, contradicting the second. So $\sim P(Q(x))$ is a theorem, and (N) gives the claim. □

Lemma F (`nup_step2`)**.** $(\Box\pi(P)\ \&\ \Box\pi(P')) \to F(Q(x))$.

Proof. Put $q := Q(x)$. We first prove

$$(\pi(P)\ \&\ \pi(P')) \to C(q). \tag{1}$$

Assume $\pi(P)$ and $\pi(P')$, and suppose, for a contradiction, that $R(q, \sim q) \vee R(\sim q, q)$. Then $F(q)$ is false: by T it would give $C(q)$, against the supposition. By the contrapositive of

Lemma D, q is true. Hence $P'(q)$ holds, and so does $P(\sim q)$, because $\sim\sim q = q$ and $F(\sim q) = F(q)$ (Lemma C). There are two cases.

Case 1: $R(q, \sim q)$. Instantiate its quantifier at $Y := P$, which is pure by $\pi(P)$. Since $P(\sim q)$, we get $\Diamond P(q)$.

Case 2: $R(\sim q, q)$. Instantiate at $Y := P'$, pure by $\pi(P')$. Since $P'(q)$, we get $\Diamond P'(\sim q)$. By Lemma C, $P'(\sim q) = P(\sim\sim q) = P(q)$, so $\Diamond P(q)$.

In both cases $\Diamond P(q)$, that is, $\Diamond P(Q(x))$, which contradicts Lemma E. This proves (1). Since (1) is a theorem of H + PE, (N) and K give $(\Box\pi(P) \,\&\, \Box\pi(P')) \to \Box C(q)$, and $\Box C(q)$ is $F(Q(x))$. □

We now use the premises. The following lemma is the only place where PP, Logical Purity, Application and Exhaustion enter.

Lemma G (`nup_step3`). *From* PP_u *and W1–W9:* (a) $\Box\pi(P)$ *and* $\Box\pi(P')$*, hence* $F(Q(x))$ *for every* x*;* (b) *for every* Y*, if* $\pi(Y)$ *then* $\pi(Y \circ Q)$.

Proof. (a) By W1 and PP_u, both M and π are pure, so W4 gives $\mathrm{Pure}(M(\pi))$, which is $\pi(P)$ by β-conversion. Since π and P are pure, W6 gives $\mathrm{Pure}_t(\pi(P))$. Now W8 applies: $\pi(P)$ is pure and true, so $\Box\pi(P)$. The same argument with W2 and W9 gives $\Box\pi(P')$. Lemma F then gives $F(Q(x))$ for every x.

(b) By W3, PP_u and W5, $M''(\pi)$ is pure, and by β-conversion it is $\lambda Y.\, Y \circ Q$. If $\pi(Y)$, then W7 gives that $(\lambda Y.\, Y \circ Q)(Y)$ is pure, and this operator is $Y \circ Q$. □

Proof of the Theorem. Assume PP_u and W1–W9, and suppose that $\mathrm{Uni}(s)$. Put $e := Q(s)$.

1. By Lemma G(a) at $x := s$, $F(e)$. By Lemma C, $F(\sim e)$, and so $Q(\sim e) = \sim e$ by Lemma B.

2. We show $R(e, \sim e)$. Let Y be an operator with $\pi(Y)$ and $Y(\sim e)$. By Lemma G(b), $Y \circ Q$ is pure. By β-conversion $(Y \circ Q)(\sim e)$ is $Y(Q(\sim e))$, which is $Y(\sim e)$ by Step 1 and Leibniz's law, so $(Y \circ Q)(\sim e)$ holds. Since $\mathrm{Uni}(s)$, in particular $R(s, \sim e)$, instantiated at $Y \circ Q$, gives $\Diamond(Y \circ Q)(s)$, that is, $\Diamond Y(Q(s))$, which is $\Diamond Y(e)$. As Y was arbitrary, $R(e, \sim e)$.

3. By T, $F(e)$ gives $C(e)$, whose first conjunct is $\sim R(e, \sim e)$. This contradicts Step 2.

Hence PP_u and W1–W9 prove $\sim\mathrm{Uni}(s)$ for arbitrary s, and so $\sim\exists x\, \mathrm{Uni}(x)$. With W10 this gives $\bot$. In this derivation Necessitation was applied only to theorems of H + PE (Lemmas B, E and F), and the premises are sentences. So the deduction theorem applies, and W proves $\sim\mathrm{PP}_u$. □

Remarks

Remark (Where W comes from)**.** W lies below Bacon's theory with a fundamental proposition. Logical Purity and Application give W1–W7. Zeroary Exhaustion gives W8–W9. W10 follows from $\exists r\,\mathrm{Fun}(r)$ and unary Recombination: if r is fundamental, Y is pure and $Y(y)$, then $\lambda z.{\sim}Y(z)$ is pure and fails at y, so Recombination gives ${\sim}\Box{\sim}Y(r)$, that is, $\Diamond Y(r)$. This uses one further purity premise, Logical Purity for $\lambda W.\lambda z.{\sim}W(z)$; the Application step is W7. Hence C with Logical Purity, Application, Exhaustion, unary Recombination and $\exists r\,\mathrm{Fun}(r)$, all as ordinary premises, refutes PP_u. Purity of Fun is not used.

Remark (Exhaustion)**.** Exhaustion is used only through W8–W9, to obtain $\Box\pi(P)$ and $\Box\pi(P')$. Lettie's companion note shows that the two Persistence formulas $\pi(P) \to \Box\pi(P)$ and $\pi(P') \to \Box\pi(P')$ would serve instead. Whether PP_u is refuted with Recombination and no Exhaustion or Persistence, all as ordinary premises, is open.

Remark (Verification)**.** In Isabelle the theorem is `w_not_pp`, a closed theorem of the calculus `U_proves` (H, PE at type t, modus ponens, Generalization and Instantiation, with no rule for added premises), whose ten antecedents are W1–W10 in the order above and whose consequent is ${\sim}\mathrm{PP}_u$. Step 4 of the proof is `nup_step4`, and the derivation of W10 in the first remark is `u_fun_ExUni`. Lettie's theories are in `theories/lettie_pp/`, unmodified; this repository's checker builds them with Isabelle2026, and its audit catalog records that these theorems depend on no oracle.

Lettie's Metamath certificate `w-not-pp`, distributed with his notes and not included here, states the same theorem with $F(z)$ written as $\Box{\sim}(R(z,{\sim}z) \vee R({\sim}z,z))$, the form used in Lettie's main note. The two forms of $F(z)$ are identical propositions by PE, but the operators F, and hence the terms M, M', M'', are written differently. The file passed three independent verifiers, each built from source on 11 November 2026: `metamath.exe`, `checkmm` and `mmverify.py`. Its trace shows that `w-not-pp` rests only on the typing, substitution and conversion rules, three propositional axioms, universal instantiation, modus ponens, Generalization and PE (as a rule from a theorem $A \leftrightarrow B$). It uses no Modalized Functionality and none of the file's purity or QLN axioms. We checked the statements of the rules it uses, not the whole axiom base.